

Syarfandi "Andi" Achmad

Fullstack Developer | End-to-End Web Application Specialist Makassar, Indonesia | [+62 852-5652-2335](tel:+6285256522335) | syarfandi.achmad@gmail.com | Portfolio: syarfandi.github.io | [LinkedIn](#)

Professional Summary

Versatile **Fullstack Developer** with 4+ years of experience engineering end-to-end digital platforms — from pixel-perfect UIs to high-performance backend APIs. Expert in the **React/Next.js** frontend and **Node.js/PostgreSQL/Supabase** backend ecosystems. Proven track record delivering production-grade applications for government agencies and enterprise startups, handling millions of records with 99.9% uptime. Passionate about clean architecture, responsive design, and shipping features that delight users.

Core Technical Competencies

- Frontend:** React, Next.js (SSR, SSG, RSC), TypeScript, JavaScript (ES6+), Three.js, Framer Motion, Tailwind CSS, Responsive Design.
- Backend:** Node.js, Express.js, Fastify, Next.js API Routes, WebSocket, PM2 Process Management, Background Job Queues.
- Database:** PostgreSQL (Stored Procedures, Indexing, Partitioning), Supabase (Realtime, RLS, Edge Functions), Redis Caching.
- API Design:** RESTful Architecture, GraphQL, WebSocket (Real-time), JWT & OAuth Authentication, Rate Limiting.
- DevOps & Deployment:** Docker, GitHub Actions, Vercel, Nginx, Linux Server Administration, CI/CD Pipelines.
- Soft Skills:** End-to-End Ownership, Technical Documentation, Asynchronous Collaboration, Remote Team Leadership.

Technical Projects (Full-Stack)

- Kawan Sedarah (Blood Stock):** A **command center** dashboard that tracks **blood stock** availability across donor units in **real-time**, powered by **Supabase** as its core backend.
- Portal Data Sulsel:** A unified information system delivering **data visualization**, **monitoring**, and religious **statistical analysis** across all districts in South Sulawesi in **real-time**.
- AI Virtual Assistant:** A **Conversational AI**-powered virtual assistant featuring **Realtime Lipsync** on 3D avatars, backed by **Voice AI** and **LLM** for immersive conversational interactions.
- GetKasir Enterprise POS:** A large-scale **Enterprise** POS ecosystem with **100+ modules**, integrating **AI**-driven sales prediction, **biometric** authentication, and a scalable **cloud-native** architecture.
- PPID Kemenag Kota Parepare:** A public information disclosure portal for **PPID Kemenag** providing access to **official documents**, **performance reports**, and an intelligent chatbot powered by **Google Gemini AI**.
- OP Tracker:** A premium logbook for tracking **One Piece** watch progress without login — equipped with automatic **filler filter**, **saga & arc** navigation map, and a gamified **pirate rank** system.
- QR Scanner Presence App:** A modern **attendance solution** leveraging **QR Code** scanning for instant participant check-in, built on a **Next.js + Supabase** stack with **real-time** sync.
- Malmora (Reseller & Dropship):** A **reseller & dropship** marketplace tailored for Muslim lifestyle products, enabling anyone to sell without physical stock across **Sulawesi and Eastern Indonesia**.
- Email Server Makassar Gov:** A city-scale **Mailcow**-based email cluster serving **200+ official domains** for Makassar City Government, hardened with **SPF/DKIM/DMARC** security protocols and **high-availability** setup.
- Helpdesk Kota Makassar:** An **e-government** system for handling **citizen complaints** digitally — featuring **real-time ticket tracking**, automated escalation, and cross-department integration across **Makassar City agencies**.
- Virtual Museum Makassar:** An immersive digital museum experience with **360° panoramic** tours, interactive **3D artifact** collections, and **audio-visual** narratives preserving South Sulawesi cultural heritage.
- AI-Haram VR:** A high-precision **virtual reality** simulation of **Al-Masjid Al-Haram** and **Al-Masjid An-Nabawi**, developed in **Unity** and optimized for both **VR headsets** and mobile devices as a Hajj & Umrah educational tool.
- Paralluta (Lalu Lintas):** A smart platform for **traffic management** in Makassar City — covering **road condition monitoring**, violation logging, and centralized **field officer coordination**.
- SOP Pelaporan Insiden Siber:** A **SOP** framework and **cyber incident response protocols** designed for **Makassar City Government** IT infrastructure, aligned with **BSSN** and **NIST Cybersecurity Framework** standards.

Professional Experience

Computer Specialist (Civil Servant) | Ministry of Religious Affairs (Regional Office)

2025 – Present | Makassar, Indonesia

- Led** end-to-end development of **Data Sulsel** — React analytics dashboard frontend + Node.js/Supabase backend serving 9M+ records with real-time data visualization.
- Built** responsive admin panels with complex form state management, PostgreSQL stored procedures, and RESTful APIs for multi-tenant government services.
- Automated** CI/CD pipelines with GitHub Actions, reducing deployment time from hours to minutes across staging and production environments.
- Implemented** JWT-based authentication, row-level security (RLS), and automated database migrations for secure multi-tenant architectures.

DevOps & Infrastructure Expert | Department of Communication & Informatics (Diskominfo)

2023 – 2025 | Makassar, Indonesia

- Developed** the **Helpdesk Makassar** citizen portal end-to-end — responsive React UI, ticket lifecycle management API, and cross-department routing backend.
- Built** admin dashboards and automated provisioning scripts for **Mailcow** email cluster serving 200+ government domains with monitoring and alerting.
- Engineered** the **Virtual Museum Makassar** — 360° panoramic React frontend with interactive 3D asset loading and backend content management API.

Co-Founder | GetKasir (Enterprise POS Startup)

2019 – Present | Remote/Project-based

- Designed** and built a **100+ module enterprise POS ecosystem** end-to-end — React merchant dashboard + Node.js/PostgreSQL backend handling thousands of daily transactions.
- Implemented** real-time inventory synchronization, sales analytics dashboards, and Supabase Realtime subscriptions for live merchant updates.
- Optimized** full-stack performance: React code-splitting and lazy loading on frontend, SQL query tuning and Redis caching on backend — 40% faster response times.
- Built** multi-branch data sync, webhook integrations, and background job queues (Bull/Redis) for automated merchant workflows.

SQA & DevOps Engineer | Malmora (Startup)

2020 – 2023 | Remote

- Built** full-stack e-commerce platform for reseller & dropship — React storefront + Node.js order processing backend serving Sulawesi and Eastern Indonesia.
- Implemented** product catalog UI components, shopping cart state management, and backend APIs for inventory synchronization and courier integration.
- Designed** database schemas for multi-vendor product catalogs, user role management, commission tracking, and automated order fulfillment pipelines.
- Automated** end-to-end testing and CI/CD pipelines, reducing deployment failures by 35% and accelerating feature delivery.

Secretary | Nurul Huda Mosque

2021 – 2026 | Makassar, Indonesia

- Managed** organizational administration and coordinated the 2021-2026 board of directors.
- Digitized** community asset management through custom full-stack solutions and automated reporting dashboards.

Certifications & Awards

- National Cyber Exercise #10** – BSSN (2024)
- Top 10 Winner** – Samsung Indonesia Next Apps 4.0 Gear VR Challenge (2017)
- Microsoft CityApp Appathon** – Participant & Developer

Education

- Bachelor of Computer Science (S.Kom)** | UIN Alauddin Makassar *GPA: 3.22 / 4.00*

Fullstack Expertise Keywords

React, Next.js, TypeScript, JavaScript, Node.js, Express.js, Fastify, PostgreSQL, Supabase, REST API, GraphQL, WebSocket, Three.js, Framer Motion, Tailwind CSS, Responsive Design, SSR, SSG, RSC, Server Components, JWT Authentication, OAuth, Redis, Docker, GitHub Actions, CI/CD, Vercel, Database Optimization, SQL Query Tuning, Indexing, Database Migration, Row-Level Security (RLS), Supabase Realtime, Edge Functions, Serverless Functions, Background Jobs, Bull Queue, PM2, Nginx, Linux, Full-Stack Architecture, End-to-End Development, Code-Splitting, Lazy Loading, State Management, Component Design.

Future-Ready Technical Roadmap

To stay competitive in the global Backend market, I am actively implementing:

- **Distributed Systems:** Designing highly-available, event-driven architectures with Kafka and RabbitMQ.
- **AI-Driven APIs:** Integrating AI models directly into backend pipelines for real-time intelligence.
- **Vector Databases:** Implementing Pinecone/Milvus for efficient AI similarity searches and RAG systems.
- **Edge Databases:** Utilizing distributed databases like Turso/Supabase Edge for low-latency global data access.